

Abdullah Emad Eldeen

AI & Embedded Systems Student | Machine Learning | TinyML | IoT

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PROFILE

Artificial Intelligence undergraduate at Menoufia University specializing in Intelligent Systems, with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Embedded Systems, IoT, and TinyML. Experienced in building ML-based applications and integrating AI models with ESP32-class microcontrollers and sensors. Seeking AI/ML, Embedded AI, or IoT internship opportunities.

EDUCATION

Faculty of Artificial Intelligence, Menoufia University — B.Sc. in Artificial Intelligence, Intelligent Systems Department

09/2023 – Present | Expected Graduation: 2027

TECHNICAL SKILLS

Programming:	Python, C, C++
Machine Learning:	Supervised Learning, Unsupervised Learning, Model Evaluation, Dimensionality Reduction, Feature Engineering
Deep Learning:	Neural Networks, CNNs, Transfer Learning, Model Optimization
Computer Vision:	OpenCV, MediaPipe, YOLO, Image Classification, Face Recognition
Embedded AI / IoT:	ESP32, Arduino, Embedded Systems, Microcontrollers, TinyML, MQTT, I2S, Sensors
Frameworks / Tools:	TensorFlow, PyTorch, NumPy, Pandas, Edge Impulse, TensorFlow Lite Micro, Arduino IDE, PlatformIO, ESP-IDF, FastAPI

PROJECTS

ESP32-S3 Keyword Spotting & TinyML (ESP32-S3 | INMP441 | TensorFlow Lite Micro | Edge Impulse | MFCC | I2S)

- Developed an embedded keyword-spotting pipeline using an ESP32-S3 and INMP441 I2S microphone.
- Implemented 16 kHz audio acquisition and integrated microphone data with an on-device inference pipeline.
- Worked with MFCC-based audio preprocessing and quantized TensorFlow Lite Micro models for speech command classification.
- Debugged audio formatting, bit alignment, preprocessing consistency, and inference issues between training and embedded environments.

Smart Adaptive Environment Monitoring System (ESP32 | MQTT | Python | Machine Learning | IoT)

- Developed an AI-enabled IoT monitoring system for real-time environmental sensing and adaptive control.
- Integrated ESP32 sensor acquisition with a Python processing and monitoring layer using MQTT.
- Applied PCA, Self-Organizing Maps (SOM), RBF-based modeling, Fuzzy Logic, and Reinforcement Learning techniques for analysis and decision-making.
- Built a live monitoring and control dashboard supporting hardware and simulation environments.

Real-Time ASL Alphabet Recognition (Python | TensorFlow/Keras | MediaPipe | Computer Vision)

- Built a real-time American Sign Language alphabet recognition system that converts hand signs into text.
- Used MediaPipe for hand tracking and a custom-trained TensorFlow/Keras classification model for sign recognition.

- Integrated computer vision preprocessing with the classification pipeline for real-time inference.

Celebrity Face Recognition System (Python | FaceNet | ArcFace | FastAPI | HTML/CSS/JavaScript)

- Developed a full-stack facial recognition application for celebrity identification.
- Integrated FaceNet and ArcFace-based embeddings for face representation and recognition.
- Built a FastAPI backend and web-based frontend for user interaction and inference.

TRAINING & COURSES

Generative AI Engineering with LLMs - IBM/Coursera (2026) | Deep Learning with PyTorch - IBM/Coursera | Tiny Machine Learning (TinyML) a program offered by HarvardX, in collaboration with edX. | Introduction to Embedded Machine Learning - Edge Impulse (2026) | Machine Learning Engineer Certification Preparation - Google Cloud/Coursera

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency | **German:** Basic